[Experiment, Analysis, and Benchmark] Maintaining Prediction Models After Policy Updates: Evaluating and Verifying Monetary Input Adjustments

Wentao Wang[1,2] **Faguo Wu**[1,2,3]
[1]*School of Artificial Intelligence*
[2]*Beijing Advanced Innovation Center for Future Blockchain and Privacy Computing*
[3]*Key Laboratory of Mathematics, Informatics and Behavioral Semantics*
Beihang University, Beijing, China
wentaowang93@buaa.edu.cn, faguo@buaa.edu.cn

***Abstract*—Policy updates change benefit levels, penalty schedules, and loan limits in public services and related regulatory activities. These changes can alter the meaning of recorded amounts and affect models trained on historical data. We develop a workflow for maintaining prediction quality by adjusting benefit, penalty, or loan amounts supplied to a model, without retraining it or modifying original records. We compare three approaches: converting amounts using old and new policy tables, selecting adjustments from model decision rules, and matching adjusted data distributions to training data. For decision trees, which score records through successive conditions, we check whether transformed reference amounts preserve the original decision paths. We further test whether replacing an adjustment preserves ranking quality or limits its decline to a predefined tolerance. Our evaluation covers 350 experimental settings from five administrative data sources sharing 37 agency–year updates. In 109 of 189 tree-model settings, the check certifies identical average precision, a ranking-quality measure, without observing outcome labels; the remaining settings require further validation. When comparing 61 scaling factors, exact score reuse is faster in 11 of 18 workloads than the fastest baseline among direct inference and two input-deduplication methods, while preserving every score. The median time saved is 74 milliseconds per batch. Hash-based integrity checks, also commonly used in blockchain systems, and external timestamps document experimental plans and associated file versions for subsequent inspection and auditing. The workflow provides verifiable evidence for maintaining models after policy updates, helping practitioners check whether adjustments preserve ranking quality and reduce repeated computation when evaluating multiple alternatives.**

***Index Terms*—Reference data, model maintenance, input repair, temporal data quality, reproducible benchmarks**

I. Introduction

Public services and regulators periodically revise benefit levels, penalty schedules, and loan limits. The same recorded amount can therefore have a different meaning under the old and new policy tables. A model trained on older records may use that amount differently from what the updated policy suggests. We study maintenance when retraining is unavailable: the trained model stays fixed, and we adjust only the benefit, penalty, or loan amount passed to it. We call this an input repair. It changes the model input, not the original administrative record. This is a benchmark restriction, not an assumption about agency policy.

We compare repairs that use different sources of information. Public-rule scaling and piecewise mapping use paired amounts from old and new policy tables, which we call anchors. Tree-compatible factors preserve the model decision paths at these anchors. Distribution matching instead uses the records being scored to align their amounts or scores with historical data. Their relative performance can change with the model, prediction target and policy update. An anchor certificate verifies model behavior only at the specified reference amounts. Rankings at other inputs require separate checks. Likewise, a small AP gap may reflect nearly identical rankings or limited measurement precision. We use paired comparisons to distinguish these possibilities.

SNAP risk-assessment reports, simulation studies and public code motivate the application setting [1], [2], [3]. Tree-path verification supplies established machinery for our finite compatibility checks [4], [5], [6], [7]. The analysis spans four application areas and accounts for shared policy updates. Paired uncertainty estimates and cross-fitting distinguish apparent selection gains from gains on held-out records. Past-only transfer tests then ask whether historical choices help on later updates, while AP audits address the narrower question of bounded substitution. We evaluate a new MSHA case population under a pre-acquisition commitment; it shares policy exposure with OSHA. We then compare exact score reuse with input deduplication and use already known labels to tighten the substitution bound. These analyses reuse exposed cohorts and preserve earlier failed tests.

Several historical results motivate the MSHA study in Section X. For trees, piecewise mapping and two continuous certificates are competitive on average; several scalar and batch controls have larger pooled losses. Whether pooled mean belongs in the competitive group depends on the weighting scheme. Representation also matters: aggregate repair differ-

ences are small for smooth scorers, whereas decile trees can reverse the direction of a repair's benefit. Selection gains vary by domain. Using half the target labels provides little gain over piecewise in SNAP and mortgages but a substantial gain in OSHA. Even so, the past-only tests retain harmful transfers. The prefix audit therefore addresses a more limited decision: can a candidate replace the reference on this scored batch without changing AP beyond a declared tolerance?

II. Task and Repair Strategies

A. Prediction actions and metrics

In SNAP, the binary endpoint distinguishes overissuance or underissuance from a correct payment among sampled QC cases. Scores prioritize further review within the released cohort; they do not replace the probability sample used for official error-rate estimation. For OSHA, we evaluate citations whose inspections had closed by the retained snapshot cutoff. A positive outcome means that the current penalty is at most 90% of the initial penalty. Settlement or adjudication may produce such a reduction, so this endpoint must not be interpreted as a payment error or a fraud finding.

The primary statistic is pooled average precision (AP), which measures how well a score ranking places records with positive outcomes near the top. Higher AP is better. We group tied scores at a shared threshold. At a five-percent budget, each group receives $\lceil 0.05 n_g \rceil$ reviews. Monthly and state-month queues have different total review counts due to rounding, so comparisons are made within each queue regime. Original SNAP queues use their archived seed-based tie order; original OSHA queues use stable citation order. New experiments declare their tie rule before execution.

Each experimental cell combines a policy transition, a record cohort and a model configuration. Cells sharing an update are not independent tasks. For a cell c and strategy s, regret is

$$r_c(s) = \max_{t \in \mathcal{S}_{\mathrm{LF}}} \mathrm{AP}_c(t) - \mathrm{AP}_c(s).$$

The reference set $\mathcal{S}_{\mathrm{LF}}$ is explicitly listed with each leaderboard. The separately trained policy model is excluded from target-label-free repair rankings. We report average tie rank, maximum and median regret, and the number of cells within 0.005 AP of the best. A small maximum over evaluated cells describes those cells; it provides no bound for an unobserved update.

B. Public-rule and batch-based controls

We use source for the earlier training period and target for the later evaluation period. For paired target anchors a_j and source anchors c_j, piecewise mapping linearly interpolates (a_j, c_j), includes zero, and extends the final segment beyond the largest anchor. Ordinary scaling uses the source/target one-person maximum ratio for SNAP and the serious-maximum ratio for OSHA. Rounded scaling applies nearest-integer rounding after that scaling.

The Kolmogorov–Smirnov (KS) control minimizes the empirical distribution distance between transformed target amounts and source amounts. The population stability index (PSI) control uses 20 source-quantile bins and 10^{-6} smoothing. Mean matching minimizes the absolute difference between mean fitted source scores and mean transformed target scores. They search the same ascending 61-factor grid $\alpha_0 + 0.0005k$, $k = -30, \ldots, 30$, with smallest-factor tie breaking. Local mean matching uses historical state-month contexts, backing off to source state and then pooled source only when the preceding group is empty. Full-window pooled matching uses the entire target batch, including records that would arrive later in a sequential deployment. First-month and rolling alternatives test sensitivity to that access.

C. Finite compatibility contract

A decision tree routes a record through successive conditions to a terminal node, or leaf. An ensemble combines predictions from several trees. We check whether each adjusted reference amount reaches the same leaves as its corresponding historical amount. Let u be the fixed context features and $\ell_f(u, b)$ the vector of leaves reached by a frozen ensemble. The reference contract is

$$\ell_f(u, g(a_j)) = \ell_f(u, c_j) \quad \forall (u, j) \in \mathcal{U} \times \mathcal{J}. \tag{1}$$

Source-target anchor correspondence comes from the public policy table, independently of outcomes. SNAP uses the minimum and 17 positive maximum anchors, extending sizes above eight by the published additional-person amount. OSHA uses nine positive schedule amounts. This is a reference relation, rather than a full entitlement calculator or an inference of the record's actual household size or violation category.

Proposition 1 (Leaf-compatible scalar interval)**.** *For an axis-aligned tree ensemble with fixed context, let $I_{u,j}$ be the monetary preimage of the paths reached at (u, c_j). The positive scalars satisfying (1) are exactly*

$$(0, \infty) \cap \bigcap_{u,j} I_{u,j}/a_j.$$

For a general repair, membership $g(a_j) \in I_{u,j}$ at every reference is equivalent to the contract.

Proof. Each visited monetary split imposes one half-line constraint. The intersection along a path preserves that path, while intersection across trees preserves the leaf vector. Pulling these constraints back through the amount representation gives $I_{u,j}$. Substitute $g(a_j) = \alpha a_j$ and divide by the positive anchor. Fixed leaf vectors imply the same ensemble score at these references.

We evaluate histogram-based gradient boosting (HGB), a random forest (RF), and HGB with amounts grouped into source-period deciles. For raw HGB, the coordinate is $\log(1 + b)$ and the monetary interval has the original split endpoint convention. For the source-decile HGB, constraints are pulled back through source-only decile boundaries. For RF, inference converts feature values to float32; the implementation accounts for the real preimage of that conversion

TABLE I
STRATEGY INTERFACE. ALL LEARNED MODELS USE SOURCE LABELS. THE ADDITIONAL-FIT COLUMN DENOTES FITTING OUTSIDE THE FROZEN-SCORE REPAIR INTERFACE.

Strategy	Information used for repair	Target batch	Additional fit	Leaf check
No repair	Identity	No	No	No
Scalar / rounded scalar	Public one-person or serious-maximum ratio	No	No	No
Piecewise	Corresponding public anchors	No	No	Anchor exactness
Certificate midpoint	Public anchors, frozen tree paths, fixed contexts	No	No	Yes
Identity-preserving certificate	Same; use public ratio if interior	No	No	Yes
Composed integer certificate	Same; fixed nearest-dollar quantizer	No	No	Yes
KS / PSI	Source amounts, transformed target amounts	Yes	No	No
Pooled mean	Source and target scores on a fixed scalar grid	Yes	No	No
State-month mean	Matched historical and target contexts	Yes	No	No
Policy HGB reference	Public policy features and source outcomes	No	Source labels	No

before checking the selected factor by actual traversal. All cases use an interior midpoint when the scalar intersection is nonempty and bounded. Otherwise the method falls back to the fixed piecewise map and records the fallback. A composed quantized repair instead intersects strict-interior preimages of admissible integers, then applies the fixed quantizer.

For N references, T trees and maximum depth D, path construction costs $O(NTD)$ and is followed by scalar intersection. We retain leaf identifiers so that the witness can be checked directly. Its guarantee concerns only the finite reference set. Two compatible repairs may move off-anchor records differently; changing the target labels can then reverse their AP ordering. We therefore measure predictive utility separately from compatibility. Ben-David et al. study broader limits of adaptation without target labels [8]. The finite-anchor argument above follows directly from the reference contract.

III. DATA, ACCESS HISTORY AND EXPERIMENTAL PROTOCOL

A. SNAP

The public QC files contain edited records from the eligible reviewed population [9], [10], [11], [12], [13]. Our fixed scope is the contiguous states and District of Columbia with finite positive certified benefits. The input whitelist is `STATE`, `YRMONTH`, and `RAWBEN`. Reviewed income, household composition, error amounts and QC outcomes are excluded from input features; the review instructions explain why household composition can differ from original certification [14].

FY2020 supplies nine available source months and FY2021 only July–September. The original cohorts contain 26,262 and 9,454 records, respectively. Pandemic interruptions limit representativeness. The 194-to-234 one-person maximum change combines an annual update and the temporary FY2021 increase [15], [16]. Later annual schedules are retained with their provenance [17], [18], [19]. The FY2024 data release includes the documented corrections to its weighting variables [9], [13]. Original scores and later refits are distinguished throughout.

B. OSHA

The original transfer uses retained official DOL inspection and violation records with 2022 source and 2023 target citations, geographically scoped to 22 states without comprehensive state plans [20]. Inputs are state, issuance month and initial penalty. Current penalty and closure are outcome-only fields; current type, deletion and contest information are excluded because they can change after issuance [21].

The source has 22,512 closed citations. All 29,047 eligible target citations were scored, after which 25,705 with valid current penalties and closed inspections entered evaluation; the remaining cases are unknown for this endpoint. The recorded snapshot cutoff is November 6, 2026. Historical source-label availability at the actual 2023 deployment date is unestablished, so this is a retrospective snapshot transfer. Source and target anchors come from the official annual memoranda [22], [23].

C. Locks, refits and claim types

The original SNAP release contains the plan, code and score hashes, and a record of target opening. For OSHA, we retain score arrays, source/target extracts and the description of historical access. These records support integrity checks against the archived bytes. They do not provide the assurance of independent prospective custody. Original locks and failed gates are left intact. Whenever a model is fitted after its target outcomes have been opened, we label that fit exposed development, even if it reproduces an original score array.

The development matrix crosses four SNAP annual transitions from FY2020–FY2024 and OSHA 2022–2023 with raw HGB, RF and source-decile HGB. HGB has 150 stages, 15 leaves, minimum leaf size 30, learning rate 0.1, L2 penalty 1, seed 1107 and disabled early stopping. RF has 200 trees, minimum leaf size 30, seed 1107 and the declared square-root feature subsampling. Decile cutpoints use source amounts only. Every strategy is evaluated in every admitted cell; no target-year tuning is allowed.

The artifact labels checks by what they actually establish. REPLAY recomputes measurements from released rows or scores; SUMMARY checks aggregate files; ARITH operates on transcribed values; REF identifies claims without producing artifacts. Quantitative conclusions in this paper rely on replayable measurements. Failed computations remain failures, including those caused by unavailable packages.

D. Mortgages, tariffs and the unit of replication

HMDA uses public CFPB conventional purchase-loan records [24], [25], [26]. The endpoint is denial (action 3) versus origination/approval not accepted (1/2), an observed lender decision rather than verified default risk or decision correctness. The first Kansas probe has one positive FHFA anchor; six regional cohorts add four county-specific unit-count limits [27], [28]. Regional cohorts each select two counties before outcomes. A dedicated two-to-four-unit cohort spans all counties in the same six states and contains 11,538 evaluated 2025 records. It overlaps the regional study and adds coverage, not independent updates.

NYC TLC supplies a fourth domain [29], [30]. The endpoint is recorded no-charge/dispute payment types 3/4 versus credit/cash 1/2, a provider-recorded payment exception. Inputs are pre-tip meter fare, rate class, fixed distance bins, weekend and pickup-hour bins. A covariate-only hash sample retains at most 50,000 rows per monthly file. The 2012 change leaves the \$2.50 base unchanged but raises the tick from \$0.40 to \$0.50 and JFK fare from \$45 to \$52 [31]. The 2022 change raises these to \$3, \$0.70 and \$70 [32]; the meter-anchor ratio spread is 16.2%. Yellow August/October 2012 supports historical calibration; yellow and green November 2022/July 2023 support later evaluation. The sampling gap and fleet changes remain confounders.

The historical SNAP schedules and OSHA2024 extension use retained official tables [33], [34]. Annual FHFA county schedules cover 2020–2025 [35]. External digest commitments use RFC 3161 [36].

The exposed synthesis contains 65 SNAP, ten OSHA, 185 HMDA and 15 TLC cells: 165 tree and 110 smooth fits. These represent 13, two, five and two domain–year policy updates, respectively. All regions, fleets and models sharing an update move together in resampling. HGB and decile HGB share a learner; Logit and SplineLogit share a link. Logit uses standardized context and log-amount features; SplineLogit adds an eight-knot cubic source-quantile spline with linear extrapolation. Both use L2 logistic regression, $C = 1$, L-BFGS, tolerance 10^{-8}, maximum 1,000 iterations and convergence warnings as failures. The 39 common-endpoint SNAP refits and 40 source-bootstrap fits provide sensitivity evidence on the same policy events.

IV. Event-Aware Strategy Comparison

A. Weighting updates and measuring uncertainty

We first subtract piecewise AP from each strategy's AP on the same records. For the primary estimate, we average model/cohort cells within a domain–year update and give each of the 22 updates equal weight. The secondary estimate instead gives each of the four domains equal weight, with equal update weights inside a domain. Reporting both shows how weighting affects the conclusion and prevents the 185 mortgage cells from dominating by count. Neither estimate represents an unspecified deployment distribution. We report each strategy's coverage; integer certificates and tariff controls cover only some settings.

Exploratory pointwise 95% percentile intervals use 4,000 update resamples within domains. All cells that share a sampled update remain together. We call a difference equivalent at margin 0.0025 only when its entire interval lies inside $(-0.0025, 0.0025)$; failure to reject a difference is not enough. The available support is limited to two OSHA and two TLC updates, and adjacent years may still be dependent. Complete event-level results and leave-one-domain-out estimates show the sensitivity to that support. Section VIII separately tests predictions from this synthesis.

B. A qualified three-level comparison

a) Competitive on average.: Both the identity-preferring and midpoint certificates meet the 0.0025 equivalence criterion under each weighting scheme. Piecewise supplies a fixed reference with a direct interpretation: it maps the published target anchors to their source counterparts. Average equivalence does not imply a tie in every cell or remove the role of a certificate in assuring finite model behavior. Pooled mean is less clear. It meets equivalence under domain weighting, but its update-weighted interval extends to -0.00683, so we cannot place it unconditionally in the same group.

b) Pooled losses requiring a specific justification.: Scalar, KS and local mean perform significantly worse under both weighting schemes. For rounded scaling, the update-weighted interval is negative, while the domain-weighted interval reaches zero. Some integer-penalty cells benefit from rounding. Those cases rule out a general recommendation to discard it.

c) Large pooled losses, with domain exceptions.: Table II shows the largest losses for no repair and PSI, with SNAP accounting for much of the difference. Removing SNAP moves PSI's point estimate to $+0.000248$ and no repair's to -0.000752. Thus the pooled ranking describes the measured four-domain average, not a loss that must occur in each domain. Identity remains a useful challenger when validating a repair for a particular deployment.

For the eight common smooth-model controls, all update- and domain-weighted intervals lie within ± 0.0025 of piecewise. Tree-style finite certificates are unavailable for these smooth scorers. Individual smooth-model cells still change, including TLC and spline controls. A simple representation calculation helps: for logit $h(u, b) = \beta^\top u + \gamma \log b$, global scaling adds a constant and preserves ranking exactly. Our fitted control uses $\log(1 + b)$, so this supplies an approximate explanation for its measured predictions.

C. Sensitivity to score changes without outcome labels

Selecting "relevant" updates because raw AP is poor would condition on the outcome being explained. Instead, define exposure as mean absolute piecewise-minus-raw score change. Before this analysis, fix its cutoff to the median of 27 decade tree cells and 25 history-calibration cells. That label-free cutoff is 0.00390. Among the 140 tree cells above it, both continuous certificate intervals remain within 0.0025; pooled mean still does not establish equivalence. The release reports

TABLE II

EXPLORATORY TREE-MODEL SYNTHESIS: 165 CELLS, 22 SHARED UPDATES. AP DIFFERENCES FROM PIECEWISE ARE MULTIPLIED BY 10^3. PRIMARY ESTIMATES WEIGHT UPDATES EQUALLY; SECONDARY ESTIMATES WEIGHT DOMAINS EQUALLY AND UPDATES EQUALLY WITHIN DOMAIN. POINTWISE 95% INTERVALS RESAMPLE UPDATES WITHIN DOMAIN (4,000 DRAWS). ALL LISTED STRATEGIES COVER ALL FOUR DOMAINS.

Strategy	Equal-update ΔAP	95% interval	Equal-domain ΔAP	95% interval
Identity-preferring cert.	-0.79	[-1.65, 0.214]	0.0134	[-0.92, 0.964]
Midpoint certificate	-0.325	[-1.17, 0.65]	1.3	[0.476, 2.15]
Pooled mean	-2.6	[-6.83, 0.0105]	0.358	[-1.67, 1.92]
Scalar	-10.1	[-16.4, -4.61]	-4.54	[-7.17, -2.21]
Rounded scalar	-6.93	[-13.5, -1.68]	-3.13	[-6.43, 0.016]
KS	-10.1	[-16.3, -4.45]	-4.53	[-7.15, -2.14]
Local mean	-9.78	[-13.6, -6.72]	-3.69	[-5.64, -1.9]
No repair	-37.9	[-57.5, -18.8]	-16.6	[-25.3, -8.16]
PSI	-47.4	[-56.3, -37.2]	-19.9	[-24.3, -15.2]

each subset's events and domains; this analysis does not tune an acceptance rule.

V. MEASUREMENT NOISE AND SELECTION OPPORTUNITY

A. Uncertainty in paired performance differences

Each cell uses 1,000 paired target-cluster bootstrap draws. Clusters are states for SNAP, inspections for OSHA, census tracts for regional/Kansas mortgages, counties for multiunit mortgages, and pickup days for taxis. Original immutable row IDs join mortgage scores to tracts. Within a draw, every strategy receives the same weights. This pairing preserves the strong correlation between strategies and yields the uncertainty of their difference; the standard error of a single AP does not answer that question. The resulting intervals condition on the trained models, fixed transforms and released scores.

Across the 70 newest cells of the exposed inventory, selected-AP SE has median 0.00975. Among the 45 original held-out cells, selected-minus-piecewise SE has median 0.000426, with an approximate two-sided 5%/80%-power detectable difference of $2.80\,\mathrm{SE} = 0.00119$. Twenty-four exceed 0.001; ten have zero paired dispersion. Some comparisons lack resolution, while others are effectively identical. H11 nevertheless has only two new domain-policy events, so 45 cellwise errors cannot establish power for future-update generalization. Its original point criterion remains unsuccessful. H10's accepted-regret threshold also remains unchanged; bootstrap uncertainty does not convert a failed point criterion into a pass.

B. Cross-fitting an outcome-informed selector

We randomly divide target clusters into two halves. The highest-AP candidate on one half is compared with piecewise on the other, then the roles are reversed. We average the two gains and repeat the split 100 times. Implementations and endpoints stay fixed, and exact selection ties favor piecewise. The selector therefore sees outcome labels unavailable to deployment repairs. Each pair of directions is one Monte Carlo replicate; it is not a pair of independent experiments, nor does the procedure produce 200 independent tests.

Cross-fitting reduces the update-weighted gain from 0.00177 for the apparent best candidate to 0.000692, with

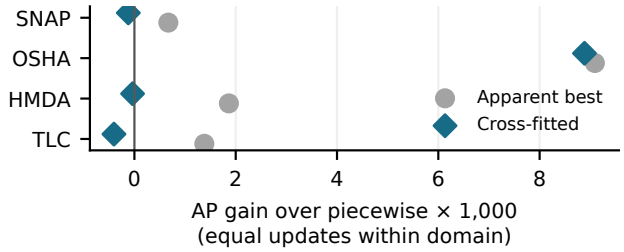


Fig. 1. Tree-model selection opportunity is domain-dependent. Points average update events within each domain. Cross-fitting uses held-out clusters; it does not bound the best achievable selector. OSHA has only two events.

event interval [0.000282,0.00106]. Under domain weighting, the corresponding gains are 0.00325 and 0.00208. These reductions are approximately 61% and 36%. They reflect winner selection, training on half the sample and differences between clusters. Attributing the entire reduction to noise would overstate what this comparison identifies. Domain results are more informative: cross-fitted gains are -0.000120 for SNAP, -0.0000360 for HMDA, -0.000404 for TLC and $+0.00889$ for OSHA. OSHA contradicts a claim that selection opportunity is universally near zero.

Two OSHA cases illustrate the remaining opportunity. For decile HGB in OSHA2023–2024, no repair beats piecewise by 0.0302 AP, and the cross-fitted selector retains 0.0302. For HGB in OSHA2022–2023, integer composition retains its 0.0206 gain. Many apparent SNAP and HMDA winners, in contrast, fail to improve AP on the held-out clusters. These comparisons help explain the limited gains observed in those domains. They do not prove that all historical selectors must fail or place an upper bound on every possible selector.

VI. MECHANISMS AND PRESERVED COUNTEREXAMPLES

a) Unchanged schedules and off-anchor behavior.: A finite compatibility witness need not be identity when the policy is unchanged. The existing unchanged-table diagnostics and all factors are retained. Fifty interior compatible factors produce 25 distinct full-target score vectors on the original SNAP cohort and 20 on OSHA. Exact declared anchors

cover 40.5% and 10.5% of their scored amounts. Anchor preservation therefore leaves a substantial off-anchor comparison problem. An explicit identity guard is appropriate for a declared unchanged regime; a midpoint alone does not supply it.

b) Discretization and integer penalties.: Ordinary versus rounded scaling changes 246 original SNAP and 11,275 OSHA scores. In the original locks, certificate-minus-rounded AP is positive in SNAP with paired interval [0.0141,0.0294], whereas rounded-minus-certificate is positive in OSHA with interval [0.00135,0.00477]. The reversal persists in 20 source-bootstrap fits per domain: rounding wins 19/20 OSHA fits and loses 20/20 SNAP fits. These fits reuse the same two updates. Decile bins supply another mechanism: off-anchor records can cross source-derived boundaries even if all reference bins remain compatible. The large OSHA2024 decile counterexample in Section V is retained.

c) Endpoint meaning before leaderboard margins.: Our post-hoc audit of the retained annual documentation found that SNAP STATUS changes from the earlier $25 error tolerance to recording errors of any amount from FY2012 [33]. We retain that original family and separately refit all 39 SNAP tree cells using AMTERR greater than $25 on both sides. No original rows are removed. The companion separates unchanged predictions under revised evaluation labels from further source-label refitting. This matters more than a small rank advantage: integer composition's narrow original SNAP advantage does not establish a portable preference under a different severity endpoint. Replacing the 39 original SNAP tree fits with these common-severity fits retains update-weighted equivalence for both continuous certificates. Under domain weighting, however, the midpoint certificate interval is [0.00116,0.00263], narrowly outside the 0.0025 margin; pooled mean now satisfies equivalence under both weightings. Thus even the competitive group is endpoint-conditional.

VII. Why Queue Qualification Did Not Establish Gain

At a fixed review budget k, let Q_s be candidate s's queue and $R(s) = \max_t |Q_t \setminus Q_s|/k$. Common records cancel, so for every binary labeling the selected queue's precision loss relative to named candidates is at most $R(s)$. The minimax-radius selector accepts only if $R \leq 0.05$ at a five-percent budget. This elementary bound is sharp for a suitable labeling and is verified in the companion. It is a candidate-relative precision statement, not an AP guarantee. A singleton has zero radius regardless of omitted alternatives.

The initial compatible-family gates look favorable partly because queues coincide. In the regional replication, two accepted Colorado cases exceed the AP-regret criterion. Admitting identity and curved anchor-exact alternatives causes complete deferral. The stronger history-qualified procedure retains candidates whose historical mean AP regret is within 0.001 of the best, then applies the fixed radius rule. Its 25 calibration cells precede 45 held-out cells; plan/code and an acquisition-only amendment received external timestamps before the corresponding downloads.

TABLE III
Original hypotheses are preserved. Full fixed thresholds, cell lists and deviations remain in the companion. Later analysis never overwrites these outcomes.

Family	Question	Outcome
H1/H2	Robustness / simple-rule losses	Fail / Fail
H3/H4/H5	Local mean / PSI / integers	Pass / Pass / Pass
H6/H7	Kansas quality / coverage	Pass / Pass
H8/H9	Regional scope / quality	Pass / Fail
H10/H11	History gate / predictive gain	Fail / Fail
H12	Multiunit / tariff coverage	Pass

Only eight of 27 held-out tree cells are accepted, including two with genuinely differing shortlisted queues and four singleton families. Colorado2024–2025 RF selects identity from a singleton and has 0.00568 AP regret. H10 therefore fails its coverage and quality conjunction. The history-qualified gain over piecewise is -0.000191 in mortgages and $+0.000195$ in taxis, yielding the H11 domain-balanced mean 0.00000222, below 0.001. Keeping every candidate accepts zero of 27 cells. The dedicated multiunit and nonproportional tariff coverage requirements H12 pass. Cross-fitting in Section V shows little held-out candidate-selection gain within these two domains, but that is supporting context, not a causal attribution or an oracle upper bound.

VIII. Separately Locked Early-Year Validation

A. Acquisition and frozen predictions

The early-year plan fixes SNAP FY2005–2009 and OSHA 2017–2021, four adjacent transitions per domain and all five unchanged model specifications. The plan, source URLs, policy tables and executable hashes received an RFC 3161 timestamp at 03:15:50 UTC on November 10, 2026, before new record acquisition. Policy/technical documentation was inspected beforehand. No earlier local SNAP records or model results for these years were located. By contrast, the earlier OSHA extraction had read full archive members before filtering later years; early OSHA is therefore raw-exposed, analysis-naive validation, excluded from the strict fresh family. This exposure distinction is fixed before new outcomes, not introduced after the results.

All five full-cohort prediction vectors are locked before deriving each target's binary endpoint and computing performance. OSHA's CSV loader can infer numeric types in outcome columns earlier; they never enter the declared features or repairs. This is computation ordering, not process-level outcome isolation. Prior targets may then become the next source. SNAP retains the documented pre-2012 $25 endpoint and uses the separate October–March and April–September FY2009 ARRA schedules [33]. OSHA retains 22 federal-enforcement states, private ownership, post-effective-date issuance windows and the same closure-conditioned endpoint. Its nine schedule anchors derive from the archived 2016

FOM and official nearest-dollar annual adjustments, with later printed gravity/max values cross-checked [37], [38]. Current snapshots still do not establish historical label availability.

B. What transfers, and what does not

P1 requires negative upper intervals for tree raw and PSI differences; P2 requires negative point differences for scalar, KS and local mean. P3 requires both continuous-certificate intervals inside ± 0.005; this wider preregistered margin does not by itself confirm the synthesis's 0.0025 margin. P4 requires cross-fitted tree selection gain no greater than 0.001. P5 requires each smooth strategy's four-update mean within ± 0.002. P6 checks the identity-preferring variant on unchanged schedules, if any. These are pointwise criteria without a familywise-error guarantee.

SNAP satisfies P1–P5. P6 is not applicable because every schedule changes. Piecewise exceeds raw by 0.0113 AP and PSI by 0.00527, while the continuous-certificate mean difference is -0.0000217. Apparent best-over-piecewise gain is 0.000179; cross-fitting gives -0.0000585. The largest absolute smooth-model mean difference is 0.00112. Seven of 12 tree cells have a feasible continuous scalar in every regime; five fall back entirely to piecewise. Across the 15 model/regime combinations, eight are feasible. The seven feasible cells also have prediction differences from piecewise, so the equivalence result is not wholly a fallback identity, but fallbacks contribute to its small average.

OSHA does not support exporting the same tiers: the corresponding P1–P4 criteria fail, P5 passes, and P6 is not applicable. Pooled mean exceeds piecewise by 0.0113 AP (interval [0.00251,0.0221]); cross-fitted selection retains 0.0159 from an apparent 0.0160. The certificate intervals no longer meet the 0.005 equivalence criterion. This is consistent with the earlier OSHA selection opportunity, not an inconvenient result to discard. The aggregate observations concern four additional updates, not independent evidence per representation.

A concrete counterexample locates the representation mechanism. On OSHA2017–2018 decile HGB, PSI changes the source-decile assignment of 5,393/29,525 evaluated records (18.3%) relative to piecewise. Exactly those prediction vectors change; AP rises from 0.714 to 0.788. The median amount difference is only \$25.2, with a selected scaling factor 0.9865 versus the public maximum ratio 0.9800. This documents a substantial ranking effect from concentration near source bins; it does not establish a causal policy effect or a rule that PSI should always be chosen. A policy-only comparison set that had discarded PSI for its pooled SNAP losses would have missed this alternative.

The full inventory now has 315 cells and 30 domain–year updates; the 275-cell synthesis and the two new 20-cell families retain separate inferential roles. All 40 new scorers are independently refitted from the retained raw inputs, with exact label/row alignment and bit-identical full-cohort predictions in the pinned runtime. The complete table, including every failed prediction and fallback, is released.

IX. Transferring Decisions and Certifying Substitutions

We test transfer across updates using the unchanged 315-cell inventory. All outcomes were exposed before this audit; its training restrictions provide retrospective internal validation rather than fresh confirmation.

a) Past-only selection.: Within each domain/scorer, average regional APs within earlier updates and weight updates equally. Select the largest historical mean after at least two updates; otherwise use piecewise. Exact ties favor piecewise. A second route minimizes maximum historical regret. The candidate support is the eight common controls, plus the two continuous certificates for trees. Task-specific integer/tariff candidates remain in their original comparisons. Training target years precede or equal the test source year. Retained snapshot labels do not establish actual historical label availability.

Past-mean selection has 22 eligible updates and 66 tree event–model decisions; 30 choose something other than piecewise. Over all 30 updates, its gain is 0.000786 AP, with interval [-0.000053,0.001709]. Domain balancing gives 0.000976 with interval [-0.000078,0.002119]. Two event–model losses exceed 0.005: OSHA2020–2021 RF loses 0.00684 and OSHA2022–2023 decile HGB loses 0.00619. Positive mean opportunity therefore fails to establish uniformly safe transfer. Leave-update-out selection looks more favorable but uses later updates, and leave-domain-out selection loses 0.00447 AP on average. Neither is a substitute for the forward test.

b) Dependence and endpoint sensitivity.: The 30 domain–year updates occupy just 19 distinct source years. Shared-calendar circular blocks of two or three observed year groups give forward-mean intervals [-0.000093,0.001675] and [-0.000087,0.001556], also crossing zero. Adjacent observed groups can span missing calendar years. These are descriptive dependence sensitivities with fixed fitted scorers and fixed selected policies; they do not establish event independence or incorporate every source of selection uncertainty. Replacing the 39 SNAP tree fits by the existing common-severity fits changes the forward mean to 0.00110 AP, with interval [0.000234,0.002063]. We retain both outcomes: endpoint definition changes the interpretation even when the transfer algorithm is fixed. No new event is created by this replacement.

c) A label-independent AP guard.: Predictive transfer and harmless substitution are different questions. For two score vectors a, b on the same n records, indexed by $i, j \in \{1, \dots, n\}$, define $A_i = \{j : a_j \geq a_i\}$ and $B_i = \{j : b_j \geq b_i\}$. For each record, these sets contain all records scoring at least as high under each method. The total variation distance between uniform distributions on these sets is

$$d_i = 1 - \frac{|A_i \cap B_i|}{\max(|A_i|, |B_i|)}, \qquad D = \max_i d_i.$$

Proposition 2 (AP substitution bound)**.** *For every binary labeling with at least one positive, tie-group average precision satisfies* $|\mathrm{AP}(a) - \mathrm{AP}(b)| \leq D$*. In particular,* $D = 0$ *certifies exact AP equality without observing any labels.*

TABLE IV

SEPARATELY PLANNED EARLY-YEAR VALIDATION. AP DIFFERENCES FROM PIECEWISE, MULTIPLIED BY 10^3, WITH FOUR-UPDATE BOOTSTRAP 95% INTERVALS. SNAP IS THE STRICT FRESH-ACQUISITION FAMILY; OSHA HAD PRIOR RAW-ARCHIVE EXPOSURE AND IS SEPARATE TEMPORAL VALIDATION. NO OUTCOMES CHANGE ANY EARLIER GATE.

Tree comparison	SNAP FY2005–2009 (4 updates, 12 cells)	OSHA 2017–2021 (4 updates, 12 cells)
No repair	-11.3 [-12.4, -9.41]	1.17 [-6.95, 15.4]
PSI	-5.27 [-7.44, -3.1]	3.77 [-4.89, 18.3]
Scalar	-2.35 [-3.46, -1.11]	1.57 [-1.46, 6.59]
KS	-2.35 [-3.46, -1.11]	2.6 [-0.816, 6.91]
Local mean	-2.68 [-3.56, -1.51]	6.08 [-0.456, 16.1]
Pooled mean	-1.01 [-2.41, -0.0856]	11.3 [2.51, 22.1]
Midpoint certificate	-0.0217 [-0.0841, 0.0248]	3.62 [-2.61, 8.04]
Identity-preferring certificate	-0.0217 [-0.0841, 0.0248]	0.446 [-15.6, 10.3]
Integer-composed certificate	0.00546 [-0.0704, 0.111]	0.379 [-15.6, 11.4]
Half-label cross-fitted selector	-0.0585 [-0.095, -0.0221]	15.9 [7.42, 27.8]

TABLE V

RETROSPECTIVE TRANSFER ON 30 UPDATES. GAINS AND INTERVALS ARE AP TIMES 10^3, RELATIVE TO FIXED PIECEWISE. MODEL/REGION CELLS ARE AGGREGATED WITHIN UPDATES. THE FIRST TWO ROUTES USE EARLIER UPDATES ONLY; THE LATTER TWO USE EXPOSED RETROSPECTIVE FOLDS.

Route	Update mean [95% interval]	Losses*
Past mean	0.786 [-0.053,1.709]	2/90
Past minimax regret	0.753 [-0.103,1.655]	2/90
Leave update out	1.264 [0.603,1.960]	3/90
Leave domain out	-4.472 [-10.190,0.252]	13/90

*Event–model outcomes with loss greater than 0.005 AP; these 90 outcomes share 30 updates. Both forward rules fall back to piecewise on the first two updates per domain (24 outcomes).

Proof. Each positive record contributes its threshold precision to AP. Its two precisions are expectations of the same binary label function under the two uniform prefix distributions, differing by at most d_i. Average over positive records and bound by D. This also covers tied scores because all positives within a tie group receive the same threshold precision. □

This bound checks whether rankings, including ties, agree. The numerical scores need not match, and agreement does not establish that the reference predicts well. A label-free $O(n \log n)$ prefix-intersection sweep computes it. Requiring both score vectors precludes an inference-cost claim.

Against piecewise, the guard certifies exact equality for the identity-preferring certificate in 109/189 tree cells, including 12 with nonidentical score arrays. The 0.005 guard accepts 110/189. The coverage is concentrated: 84 of the 109 zero cases are HMDA, whose 111 cells share five updates. Scalar has 102/189 zero cases, including 85 HMDA cases. For smooth scorers, scalar has 49/126 zero cases and 67/126 cases within the 0.005 bound. These are audit coverage counts, not independent successful tasks. The companion gives every domain and candidate, including the often-vacuous bounds. All 2,898 AP values in this common-support audit replay exactly; exhaustive small weak-order checks verify the bound independently.

d) Operational consequence.: Align endpoints and tables, then score piecewise and the proposed substitute. If the guard meets the declared AP-loss tolerance, substitution has a reference-relative guarantee on that batch. Otherwise retain the reference or collect domain-matched labels. Failure to pass the guard does not establish that the candidate is worse. Historical winners can prioritize validation, but the forward counterexamples preclude guaranteed improvement.

X. NEW-SOURCE VALIDATION AND A LABEL-BUDGET GUARD

a) A new agency, with shared policy exposure.: After the four-source analysis, we acquired MSHA assessed-violation and civil-penalty-decision archives [39]. An externally verified plan timestamp (10 November 2026, 06:53:53 UTC) precedes the first case download (06:54:18 UTC). The seven August–December transitions, 2015–2016 through 2021–2022, use all mines meeting the fixed regular-assessment filters and the same five model configurations. A positive endpoint means that the latest valid contested decision reduces the original proposed penalty by at least 10%. Noncontested or undecided records are excluded from outcome evaluation. Features are the proposed amount, coal/metal and good-faith context, and billing month. All five full target score vectors are committed before interpreting each target year's outcomes. Raw outcome bytes remain locally accessible; this is pre-acquisition planning, not independently blinded custody. A separately timestamped header alias reconciles the official field description with its shortened archive name, before any target interpretation.

The addition supplies 35 cells and seven agency–year updates, yielding 350 cells and 37 updates across five sources, still only 19 distinct source years and four application areas. MSHA and OSHA share federal inflation legislation: the addition tests a different case population and outcome mechanism, not seven new independent macroeconomic shocks. Primary historical families and their results remain unchanged.

b) Public correspondence can be many-to-one.: The 2015 regular-assessment table has distinct amounts at points 140–144, from $60,000 to $70,000; the 2016 table caps all five

at $68,300 [40], [41]. A target amount alone cannot uniquely recover the old point-dependent amount. We predeclared a midrange piecewise convention and retain lower/upper conventions and both minimum- and cap-based scalar controls. If the old amounts at a merged knot reach distinct reference leaf vectors under one context, no deterministic amount-only repair satisfies every corresponding leaf constraint. This is an immediate functional incompatibility, not a claim that every fitted model exhibits it. All 21 fitted tree cells have zero such leaf conflicts, and lower/midrange/upper piecewise conventions yield identical scores. Thus the public inverse is ambiguous while these fitted scorers are insensitive at the merged cap. In the nonproportional 2015–2016 transition, however, HGB cap-based scaling loses 0.01542 AP relative to piecewise, whereas minimum-based scaling gains 0.00038. The complete table and model sensitivity matter more than the name "scaling."

c) From worst-case ranks to a sampled-label guard.: The maximum-prefix bound in Proposition 2 is often loose. Sort its recordwise values as $d_{(1)} \geq \cdots \geq d_{(n)}$ and define $G(p) = p^{-1} \sum_{i=1}^{p} d_{(i)}$ for integer $p \in \{1, \ldots, n\}$.

Proposition 3 (Finite-population label-budget guard)**.** *If a fixed batch contains $P \geq 1$ positives, then $|\,\mathrm{AP}(a) - \mathrm{AP}(b)| \leq G(P)$. Let L be a one-sided $(1-\alpha)$ lower confidence bound on P obtained from a simple random sample without replacement. With probability at least $1-\alpha$, all fixed candidate pairs on that batch simultaneously obey their respective $G(\max(1, L))$ bounds. A selector may use the sampled labels to choose among candidates satisfying a declared tolerance.*

Proof. The AP difference is at most the mean of d_i over the P positive records. Their sum is at most the sum of the P largest values. Since G is nonincreasing, $L \leq P$ implies the claimed inequality for every fixed pair. This one event is shared by all pairs, including an adaptively selected pair. If $P = 0$, both APs use the zero convention. Exact hypergeometric inversion supplies L without assuming independent record labels. □

This combines elementary finite-population inference and the prefix bound; it is not a new general concentration theorem. Candidate scores must be fixed before label sampling. The guarantee is relative to piecewise on this batch, not to an unknown optimal method or a future population. Requiring both score vectors rules out first-pass inference savings.

We freeze budgets 0, 50, 200 and 1,000 labels, 100 nested random samples, a 0.005 AP tolerance, and piecewise-first ties. At positive budgets the selector maximizes pilot AP among admitted candidates. For each repetition, $\alpha = 0.05/(3 \cdot 7)$ controls all seven cohorts, three positive budgets, and all fixed model/candidate pairs jointly; it does not make a joint claim over the 100 repetitions. Utility is full-cohort AP, including the pilot. These experiments simulate revealing random subsets of real, retained labels; they are not new human annotations. Repetitions are label-sampling variability, not new tasks.

At 200 labels, the predeclared extra-coverage target (at least four events) passes; the separate predictive-utility target (gain at least 0.001 with a positive interval) fails. These are distinct criteria: certifying a safe replacement need not identify a better predictor. On the exposed 315-cell inventory, extra coverage appears in 19/30 tree update groups at this budget, but mean selected gain is only 0.00000141 AP. Eight of 21 tree cells meet the 80-percent additional-coverage criterion; seven are RF. The 200 labels represent 3.2–5.8 percent of these evaluated cohorts. We retain all four budgets, both representations, all sampling failures and every original hypothesis.

TABLE VI
NEW MSHA VALIDATION. EXTRA EVENTS: AT LEAST ONE TREE MODEL ADMITS AN ADDITIONAL CANDIDATE BEYOND THE ZERO-LABEL GUARD IN AT LEAST 80% OF SAMPLES. GAIN AND INTERVAL ARE EVENT-MEAN AP TIMES 10^3; SEVEN EVENTS, FIXED SCORERS.

Labels	Extra events	Gain [95% interval]	Losses*
0	0/7	0.000 [0.000, 0.000]	0
50	7/7	0.000 [-0.000, 0.001]	0
200	7/7	0.002 [-0.003, 0.007]	0
1,000	7/7	0.008 [-0.002, 0.022]	0

*Selected loss greater than 0.005 in 2,100 tree cell–sample outcomes per budget; these outcomes share seven events.

A post-hoc opportunity audit explains the failed gain target: the fixed-candidate oracle gap is 0.002014 AP, but the best admitted candidate averages only 0.00001577. The conservative guard excludes most observed opportunity before pilot selection. This is an in-sample descriptive decomposition, not an oracle bound on achievable transfer or a causal attribution to noise.

d) An executable decision procedure.: First inspect whether public anchor correspondence is single-valued. If it merges incompatible old behaviors, retain the ambiguity or obtain the missing context instead of claiming an exact inverse. Next compare the reference and candidate rankings: $D \leq \delta$ permits a label-free, reference-relative substitution. If this fails and a label budget is available, use the sampled-count guard with a prespecified simultaneous confidence level. A passed guard bounds possible loss; a failed guard calls for further validation and does not prove inferiority. Empirical winners guide which alternatives to inspect, while the guard answers whether the proposed substitution is defensible on the current batch.

XI. PRACTICAL VALUE: EXACT REUSE AND A TIGHTER AUDIT

Small AP gains do not demonstrate lower maintenance cost. We therefore ask whether a certificate can reduce the computation needed to score repeated repairs while preserving every prediction. This engineering study was designed after review and uses exposed cohorts. It contributes neither new policy events nor blind outcome tests.

a) An exact reuse contract.: Fix the complete nonmonetary model input u. At reference monetary coordinate z_0, intersect all visited tree-path inequalities in the native inference coordinate to obtain $I_u(z_0) = (l, h]$. Every finite z in this interval reaches the same leaf vector, so the unchanged

TABLE VII
COLD 61-FACTOR SCORING PLUS FEATURE PREPARATION, SECONDS. FASTEST CONTROL → CERTIFIED REUSE; EACH ENTRY ADDS MEDIAN PREPARATION AND MEDIAN SCORING (THREE RUNS PER STAGE). ALL 18 SCORE MATRICES ARE EXACTLY PRESERVED.

Cohort	HGB	RF	Decile
HMDA CA23–24	0.35→0.28	0.32→0.32	0.23→0.23
MSHA17–18	1.00→0.92	0.90→0.91	0.94→0.80
MSHA21–22	1.13→1.01	0.96→1.15	0.92→0.78
OSHA23–24	3.29→3.18	4.11→4.89	2.01→2.08
SNAP20–21	2.14→0.73	2.71→2.16	0.31→0.37
TLC22–23	2.34→1.68	2.38→2.13	1.28→1.30

deterministic ensemble returns exactly the same score. RF coordinates are cast to float32 before checking membership; HGB uses log amounts and decile HGB uses integer bins. Different contexts never share entries. The implementation constructs entries at the central repair, looks up other grid values, and scores every uncovered input with the unchanged model. Consequently the entire candidate-score matrix, all rankings and all label-dependent decisions are preserved, without acquiring labels. This is a direct use of path compatibility, not a new general inference algorithm.

b) Cold-cache comparison against input deduplication.: We fix six existing cohorts from all five sources, three tree representations, grid sizes $K \in \{1, 5, 15, 61\}$ and three randomized-order timing repeats. The MSHA2015–2016 development pilot is excluded. Every trial rebuilds its cache. We include complete-context grouping, feature preparation, interval construction, fallback inference and materialization of the full output matrix. Model fitting and data loading are common and excluded. Two exact-input controls use either global structured-key deduplication or faster per-context scalar sorting with batched prediction; direct scoring is also retained. Each reported comparison uses the fastest median among these three controls, adding separately measured median preparation and median scoring times. Timing repeats are not new scientific tasks. Restricting the input features creates repeated contexts. With more features or mostly unique contexts, constructing reuse intervals may cost more than it saves.

For the 61-factor workload, reuse has a lower median time in 11/18 cells; its geometric-mean speed ratio is 1.11, with range [0.83,2.95]. A ratio below one is a slowdown. A descriptive 5-percent margin leaves 9/18 median improvements, and 8/18 exceed that margin in every timing repeat; these are noise sensitivities, not confidence statements. For a single repair, it is faster in 0/18 cells. Interval construction therefore needs repeated scoring to offset its cost. Ordinary deduplication remains a necessary baseline. All retained 61-factor matrices reproduce bit-identically. The median absolute saving against the fastest control is 0.074 seconds per 61-factor batch. These host-conditioned measurements cover the scoring stage, not agency labor, total pipeline latency or deployment cost; no predictive improvement is claimed. The saving applies when comparing many repairs. A maintainer who already uses one fixed piecewise repair can avoid this grid entirely; faster grid evaluation alone is not a reason to adopt batch matching.

c) Use the identities of already queried labels.: The preceding bound uses the number of positive sampled records but ignores their identities. Incorporating which records are positive can tighten it. For a known set S, let $c = \sum_{i \in S} y_i$, $w = \sum_{i \in S} y_i d_i$, and let $u_{(j)}$ be the descending d_i values among unknown records. The maximization below ranges over integer counts k. On the same event $P \geq L$, a tighter bound is

$$B_S(L) = \max_{\substack{0 \leq k \leq n-|S| \\ c+k \geq \max(1,L)}} \frac{w + \sum_{j=1}^{k} u_{(j)}}{c+k}. \tag{2}$$

For the actual unknown-positive count k, their contribution is bounded by the largest k unknown values. Maximizing over feasible counts proves validity; every feasible numerator is also bounded by the largest $c + k$ values over all records, so $B_S(L) \leq G(\max(1, L))$. If a census finds no positives, both APs are zero. The simultaneous statement uses the identical prevalence event, without an additional multiplicity penalty. Exhaustive known-label patterns verify this envelope; it is tight for this rowwise relaxation, not necessarily for the actual AP difference.

Keeping all 35 MSHA cells, the complete candidate families, the same 100 samples of 200 labels, the same confidence level and the same 0.005 tolerance, the refined bound admits extra candidates in 9/21 tree cells at 80% sampling probability (previously 8/21). The admitted in-sample oracle changes from 0.00001577 to 0.00001580 AP; the full-family oracle remains 0.00201403. Selected mean gain changes from 0.00000198 to 0.00000200. This exposed-data sensitivity does not replace the failed original gain target. The remaining gap measures conservatism of the audit, not absence of useful candidates.

d) A decision rule with separate evidence requirements.: When scoring a single repair, begin with direct inference or exact-input deduplication. If many repairs alter the same coordinate, measure cold-cache cost against both controls on the actual batch. Use interval reuse only when the repeated work pays for construction. Its exact prediction-preservation property does not depend on target-label custody. Replacing predictions is a different decision: the known-label guard can bound reference-relative loss, but a separately measured benefit is still needed to justify the replacement.

XII. LIMITATIONS AND PRACTICAL USE

The historical synthesis and past-only transfer audit are exposed, exploratory evidence; the MSHA addition follows a pre-acquisition plan with locally accessible outcomes. The historical 30 updates occupy 19 observed source years; the added seven MSHA transitions share existing years and federal policy exposure. These are dependent groups rather than independent randomized tasks. Neither weighting scheme makes these policies representative. HMDA's many cells share just five almost proportional annual updates, with purposive two-county regional cohorts. The multiunit cohort improves coverage but overlaps those regions. TLC contributes only two

tariff changes, with long sampling gaps and provider-recorded payment exceptions. SNAP QC selection, pandemic gaps and endpoint changes restrict interpretation. OSHA conditions on inspection closure and uses current snapshots, so source labels may not have been available at the historical deployment date.

No result estimates a causal policy effect, monetary savings, fairness or agency adoption. Section XI measures only repeated-scoring runtime under the released backend; its eighteen workloads reuse six existing cohorts. Deterministic score preservation does not require independent labels, whereas predictive-transfer claims still do. We report these evidence types separately.

A hash identifies the retained file bytes, and a timestamp records that the digest existed by a given time. These records allow later integrity checks; they do not establish independent label custody or rule out access elsewhere. The three older strict RF AP replay failures remain documented; serial inference corrects their numerical cause in separate refits without changing the old locks. Their AP changes are at most 0.00004 and preserve the original queue/gate decisions. All 70 later history-study fits reproduced bit-identical predictions. New analyses preserve paired score uncertainty but do not incorporate refitting the source model or re-estimating every transductive transform on resampled records. Exact numerical replay is conditional on the pinned runtime.

A maintainer should begin by checking the endpoint and the tables' effective dates, then examine the scorer's representation and measured exposure. For trees, a practical comparison includes piecewise, both continuous certificates and identity, with additional challengers justified by domain evidence. OSHA warrants integer composition, pooled mean and PSI: excluding them solely because of SNAP performance would discard large observed gains. Taxi comparisons should retain direct tariff inverses. Near-proportional anchors alone do not validate scalar or distribution matching. For smooth scorers, first determine whether the update changes rankings enough to justify selection. A queue radius concerns the named queues only. Likewise, the AP guards in Sections IX and X bound substitution relative to a reference on the current batch. Evidence that the substitute improves prediction must come from domain-appropriate outcome validation.

XIII. Related Work

SNAP risk-assessment systems and public simulation studies motivate policy-aware scoring [1], [2], [3]. Our benchmark isolates maintenance of a frozen scorer; it does not evaluate these complete review workflows or establish superiority over them. Sparse coordinate-wise adaptation preserves a pretrained source model [42]; keypoint-guided optimal transport uses cross-domain correspondences [43]. Affine statistics transport considers frozen-model adaptation [44], while TransferScore evaluates adaptation models without target labels [45]. We use public effective-dated anchors and an explicit finite model contract.

Prior work documents online-batch dependence [46] and sensitivity to label-distribution shifts in test-time normalization [47]. For these administrative cohorts, we examine batch sensitivity through source-context matching and rolling and first-month controls. Tree path constraints and ensemble verification supply the certificate's elementary machinery [4], [5], [6], [7]. Efficient tree inference is established: QuickScorer optimizes ensemble traversal [48], and TL2cgen compiles models into native code [49]. Our cold-cache study tests workload-specific reuse under the released backend against exact-input deduplication; it does not claim a new general inference engine or superiority to compiled backends. CurrentClean and HoloClean address stale or inconsistent values with richer learned evidence [50], [51]; a scoring-coordinate repair here makes no assertion that a benefit payment or penalty itself has been corrected.

XIV. Conclusion

Input-adjustment evaluation asks three questions: are policy-anchor predictions preserved, is substitution safe on this batch, and does prediction improve after a new update? Compatibility follows from model structure. AP bounds can certify substitution with no labels or with a finite-population label sample. Improvement on a new update still requires outcome validation. Event-weighted synthesis and early-year replay expose domain and endpoint exceptions to simple strategy rankings. Past-only selection delivers small average gains yet retains harmful transfers, while exact ranking equivalence certifies a substantial, domain-concentrated subset without labels. The MSHA sequence preserves shared-policy dependence and keeps safe substitution separate from improved prediction. Tree-path reuse preserves scores exactly, but saves computation only when repeated scoring offsets interval construction. Using the identities of known labels tightens the substitution bound; it does not establish a predictive benefit.

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